

CHURN DATA ANALYSIS

1. Executive Summary

This project presents a customer churn data analysis workflow implemented in Python. The supplied notebook connects a SQLite customer churn database to Python using sqlite3 and pandas, imports database tables using SQL queries, performs data cleaning and quality checks, combines customer, subscription, and support information, engineers analytical features, and performs exploratory data analysis (EDA). The analysis also uses Matplotlib and Seaborn to visualize churn trends, plan-level churn, correlations, and multidimensional relationships.

2. Objectives

- Connect a SQL/SQLite database to Python using sqlite3 and pandas.
- Import database tables and data into pandas DataFrames using SQL queries.
- Clean and standardize customer, subscription, and support datasets.
- Perform data-type conversion, column selection/removal, renaming/standardization where required, and data-quality checks.
- Handle missing/null values using appropriate mapping and filling techniques.
- Create calculated columns and transformed features using NumPy and pandas.
- Combine multiple datasets using customer ID as the common key.
- Perform customer churn analysis using aggregation, groupby operations, and pivot tables.
- Create visualizations using Matplotlib and Seaborn to communicate analytical results.

3. Technology Stack

Technology / Library	Purpose
SQLite / sqlite3	Database connection and SQL table creation/access
SQL	Querying database metadata and importing table data
Pandas	Data loading, cleaning, transformation, merging, aggregation and EDA
NumPy	Conditional feature engineering and numerical operations
Matplotlib	Line and bar chart visualization
Seaborn	Heatmap, pairplot and categorical/multidimensional visualization
Jupyter Notebook	Interactive development and analysis

4. Overall Data Analysis Workflow

1. Connect to the SQLite database.
2. Identify available database tables using SQL.
3. Import each table into pandas DataFrames.
4. Inspect schema and column information.
5. Clean and standardize individual datasets.

6. Convert date fields to datetime format.
7. Handle missing country values using state-to-country mapping.
8. Create the churn flag from cancellation status.
9. Calculate complaint counts and retain the relevant customer-level support record.
10. Merge subscription, customer and support datasets using customerid.
11. Engineer tenure, churn risk and escalation-related features.
12. Calculate churn, retention, revenue, tenure and complaint KPIs.
13. Perform groupby, aggregation and pivot-table analysis.
14. Create Matplotlib and Seaborn visualizations.
15. Export the consolidated dataset to CSV.

5. SQL Database Connection and Data Import

5.1 Connecting SQLite to Python

The notebook uses `sqlite3` to establish a connection with the `customer_churn.db` database. Pandas `read_sql` is then used to execute SQL queries and return query results as DataFrames.

```
conn = sqlite3.connect("customer_churn.db")
```

5.2 Discovering Tables with SQL

The notebook queries `sqlite_master` to identify tables stored in the SQLite database. For each discovered table, `SELECT *` is executed through `pandas.read_sql` and the resulting data is stored as a DataFrame.

```
sql_query = """
    select name
    from sqlite_master
    where type = 'table'
"""
tables = pd.read_sql(sql_query, conn)

for table_names in tables['name']:
    df = pd.read_sql(f"select * from {table_names}", conn)
```

5.3 Schema Inspection

The notebook also uses `PRAGMA table_info(table_name)` to inspect the columns of each SQLite table. This supports understanding the database schema before analysis.

6. Data Cleaning and Quality Checks

Data cleaning is performed separately on the customer, subscription and support datasets before they are merged.

6.1 Column Selection / Removal

Columns that are not required for the analysis are removed. In the customer data, interests and pincode are dropped. In the support data, `col_1` and comment are removed.

```
df_db_customer.drop(columns=['interests', 'pincode'], inplace=True)
df_db_support.drop(columns=['col_1', 'comment'], inplace=True)
```

6.2 Data-Type Conversion

Date-related columns are converted to pandas datetime format. This is necessary for date arithmetic, tenure calculation and monthly churn analysis.

```
df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])

date_type = ['cancellation_date', 'renewal_date', 'subscription_start_date']
df_db_subscription[date_type] = df_db_subscription[date_type].apply(pd.to_datetime)
```

6.3 Standardizing Categorical/Text Values

Gender labels are standardized from Men/Women to Male/Female, and customer names are converted to title case.

```
df_db_customer['gender'] = df_db_customer['gender'].replace({'Men': 'Male', 'Women': 'Female'})
df_db_customer['name'] = df_db_customer['name'].str.title()
```

6.4 Missing / Null Value Handling

Missing country values are filled using a state-to-country mapping derived from records where country is already available. The notebook then checks remaining records where country is still missing.

```
state_country_mapping =
df_db_customer.dropna(subset=['country']).set_index('state')['country'].to_dict()
df_db_customer['country'] =
df_db_customer['country'].fillna(df_db_customer['state'].map(state_country_mapping))
```

6.5 Data Quality Checks

- DataFrame information is inspected using `info()` for customer, subscription and support datasets.
- Unique customer counts are checked for each source dataset.
- Remaining missing country values are explicitly inspected.
- DataFrame shape and `head()` are used after merging to verify the resulting dataset.
- Column names are inspected before downstream calculations.

7. Feature Engineering

Feature engineering creates analytical columns from existing fields. The notebook uses both pandas and NumPy for this stage.

7.1 Churn Flag

A binary churn flag is generated from `cancellation_date`. Customers with a cancellation date are assigned 1; otherwise 0.

```
df_db_subscription['churn_flag'] = np.where(df_db_subscription['cancellation_date'].notna(), 1, 0)
```

7.2 Complaint Count

The number of support records for each customer is calculated using groupby and transform.

```
df_db_support['complaint_count'] = df_db_support.groupby('customerid')['customerid'].transform('count')
```

The support data is then sorted by complaint_count and duplicates are removed by customer ID, keeping the record associated with the highest complaint count.

7.3 Tenure

Customer tenure is calculated in days. For churned customers, tenure is measured from subscription start to cancellation. For active customers, it is measured from subscription start to the current date used when the notebook runs.

```
today = pd.Timestamp.today()

df['tenure_days'] = np.where(
    df['cancellation_date'].notna(),
    (df['cancellation_date'] - df['subscription_start_date']).dt.days,
    (today - df['subscription_start_date']).dt.days
)
```

7.4 Escalation Transformation

The escalation field is transformed from Y/N into a binary numerical feature so that correlation and numerical analysis can be performed.

```
df['escalations'] = np.where(df['escalations'] == 'Y', 1, 0)
```

7.5 Churn Risk Category

A churn risk category is derived from churn_score using NumPy select: scores below 50 are low risk, 50–69 are medium risk, and 70 or above are high risk.

```
cond = [
    (df['churn_score'] < 50),
    (df['churn_score'] >= 50) & (df['churn_score'] < 70),
    (df['churn_score'] >= 70)
]
state = ['low', 'med', 'high']
df['churn_risk'] = np.select(cond, state, default='unknown')
```

8. Data Integration / Merging

The cleaned datasets are integrated through customerid. The subscription dataset is used as the starting DataFrame, followed by left merges with customer and support information.

```
df = (df_db_subscription
      .merge(df_db_customer, on='customerid', how='left')
      .merge(df_db_support, on='customerid', how='left'))
```

After merging, the notebook checks the resulting shape, previews the data, and examines customer counts across the source tables.

9. Data Analysis and Exploratory Data Analysis

The notebook calculates business-oriented customer churn metrics and uses groupby and aggregation operations to compare customer segments.

9.1 Key Performance Indicators

KPI	Calculation / Meaning
Churn Rate	Mean of churn_flag × 100
Retention Rate	100 – churn rate
Average Monthly Charges	Mean of monthly_charges
Average Tenure	Mean of tenure_days
Revenue Loss	Sum of monthly_charges for churned customers
Escalation Rate	Percentage of records with escalation = Y
Average Customer Complaint	Total complaint count divided by unique customers

9.2 Groupby Analysis

Groupby operations are used to examine churn and revenue-related measures across plan type, state and subscription type.

```
df.groupby('plan_type')['churn_flag'].mean().mul(100)
df.groupby('subscription_type')['churn_flag'].mean().mul(100)
df.groupby('state')['monthly_charges'].sum()
```

9.3 Correlation Analysis

The notebook calculates the correlation between the binary escalation feature and churn_flag after removing missing values from the selected fields.

```
correlation = corr_df['escalations'].corr(df['churn_flag'])
```

9.4 Pivot Table Analysis

Pivot tables summarize churn rate, monthly charges and unique customers by plan type.

```
pd.pivot_table(
    df_visual,
```

```

values=['churn_flag', 'monthly_charges', 'customerid'],
index='plan_type',
aggfunc={
    'churn_flag': 'mean',
    'monthly_charges': 'sum',
    'customerid': 'nunique'
}
).reset_index()

```

10. Data Visualization

The notebook uses Matplotlib and Seaborn to communicate patterns identified during the analysis.

10.1 Matplotlib

- Monthly churn trend: a line plot showing the number of churned customers by cancellation month.
- Plan trend: a bar chart comparing average churn_flag across plan types.

```

plt.figure(figsize=(9,3))
plt.plot(churn_trend.index.astype('str'), churn_trend.values, marker='o')
plt.title('Churn Monthly Trend')
plt.ylabel('Churned Customers')
plt.xlabel('Monthly')
plt.show()

```

10.2 Seaborn

- Correlation heatmap for selected encoded categorical and numerical variables.
- Pairplot for multidimensional relationships among selected variables.
- Categorical plot comparing monthly charges by plan type, gender and churn-risk category.

```

sb.heatmap(df_encoded.corr(), annot=True)

sb.pairplot(df_encoded)

sb.catplot(
    data=df_visual,
    x='plan_type',
    y='monthly_charges',
    hue='gender',
    col='churn_risk'
)

```

11. Data Export

The consolidated churn dataset is exported from pandas to CSV for reuse outside the notebook.

```
df.to_csv('exported_churn_data.csv', index=False)
```

12. Analysis Outputs

- A consolidated customer churn DataFrame is created from subscription, customer and support datasets.
- The workflow produces churn and retention KPIs.
- Customer tenure, complaint count, escalation and churn-risk features are engineered.
- Churn can be examined by plan type and subscription type using groupby operations.
- Revenue-related measures such as average monthly charges and revenue loss from churned customers are calculated.
- Pivot tables provide summarized customer, revenue and churn information by plan type.
- Matplotlib and Seaborn visualizations provide trend, comparison, correlation and multidimensional views.

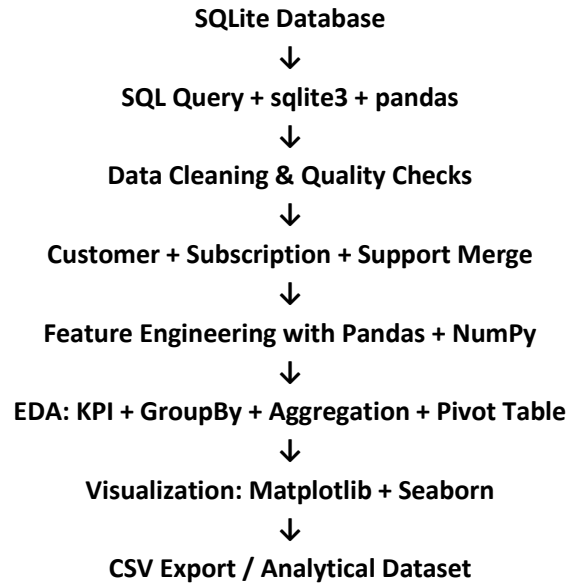
13. Skills Demonstrated

Area	Skills Demonstrated
SQL + Python Integration	sqlite3 connection, SQL queries, pandas.read_sql
Data Import	Importing SQLite tables into pandas DataFrames
Data Cleaning	Data types, column removal/selection, text standardization, null handling
Data Quality Checks	info(), unique counts, shape/head checks, missing-value inspection
Feature Engineering	Calculated columns, conditional logic, transformations, filters
NumPy	np.where, np.select and numerical calculations
Pandas	merge, groupby, aggregation, pivot_table, datetime operations
EDA	KPI calculation, segmentation, aggregation and correlation
Visualization	Matplotlib line/bar charts and Seaborn heatmap/pairplot/catplot
Data Export	CSV export using pandas

14. Conclusion

The project demonstrates an end-to-end Python-based data analysis workflow starting from a SQLite database and ending with a cleaned, integrated and analysis-ready customer churn dataset. The notebook covers the major stages of practical data analysis: SQL database connectivity, data import, cleaning, quality checks, feature engineering, exploratory analysis, aggregation, pivot-table analysis and visualization. The resulting workflow provides a structured foundation for understanding customer churn patterns and preparing the dataset for further reporting or predictive analysis.

15. Project Workflow Summary



16. Results and Findings

The following results are taken directly from the supplied Jupyter Notebook outputs. They represent the 21-customer dataset processed in the notebook.

16.1 Dataset Results

Dataset / Result	Output
Customer records	21
Subscription records	21
Support records	9
Unique customers with support records	7
Final merged DataFrame	21 rows × 21 columns
Final engineered DataFrame columns	22
Remaining missing country values	0

16.2 KPI Results

Metric	Result
Churn Rate	28.57%
Retention Rate	71.43%
Average Monthly Charges	18.85
Average Tenure	1535.14 days
Revenue Loss from Churned Customers	73.94
Escalation Rate	14.29%
Average Customer Complaint	0.43
Escalation vs Churn Correlation	0.65

16.3 Churn by Plan Type

The notebook calculated churn percentage by plan type. The results show the churn rate for each plan in the analyzed dataset.

Plan Type	Churn Rate
Basic	60.00%
Premium	14.29%
Standard	22.22%

16.4 Subscription Type Analysis

Subscription Type	Reported Percentage
Organic	0.00%
Paid	16.67%
Refferal	83.33%

These percentages reproduce the notebook's subscription_type groupby output. They should be interpreted according to the metric calculated in the notebook rather than as a separately validated causal relationship.

16.5 Plan-Level Pivot Table

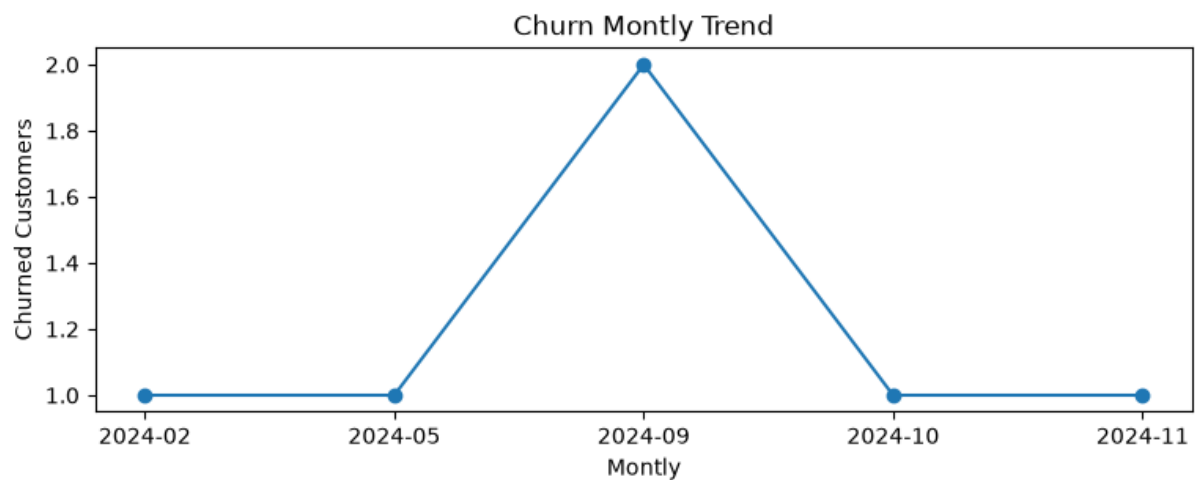
Plan Type	Churn Flag Mean	Customers	Monthly Charges
Basic	0.6000	5	52.95
Premium	0.1429	7	218.93
Standard	0.2222	9	123.91

The churn_flag mean corresponds to the churn proportion for each plan. The customer column represents the number of unique customers and monthly_charges is the aggregated value produced by the pivot table.

17. Graphs and Visual Results

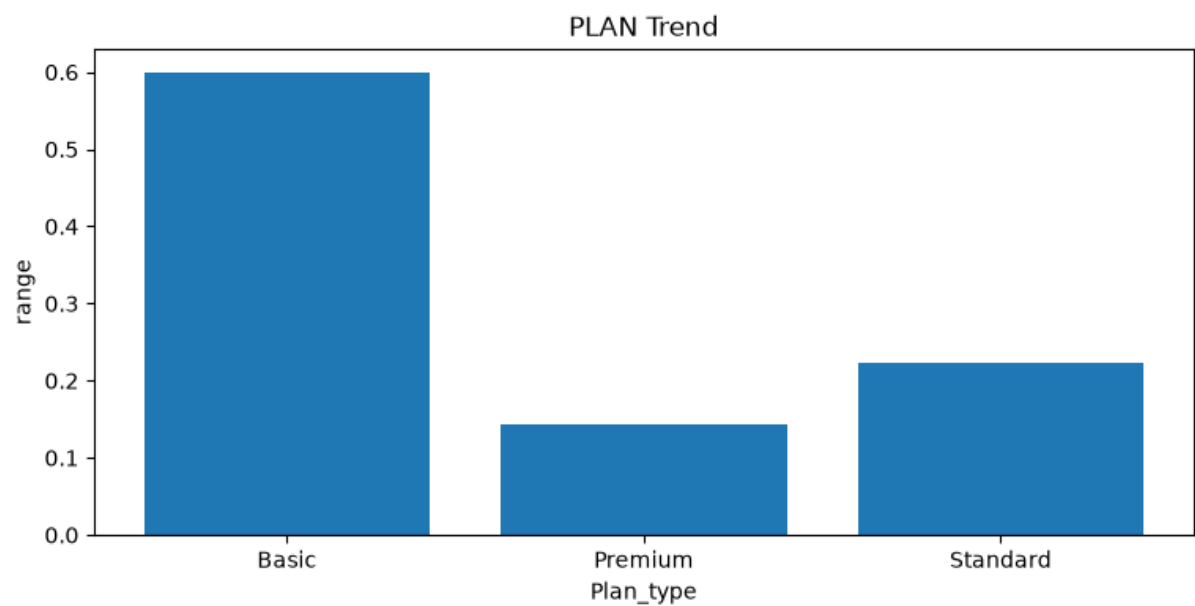
17.1 Monthly Churn Trend

This line chart visualizes the monthly trend of churned customers based on cancellation month.



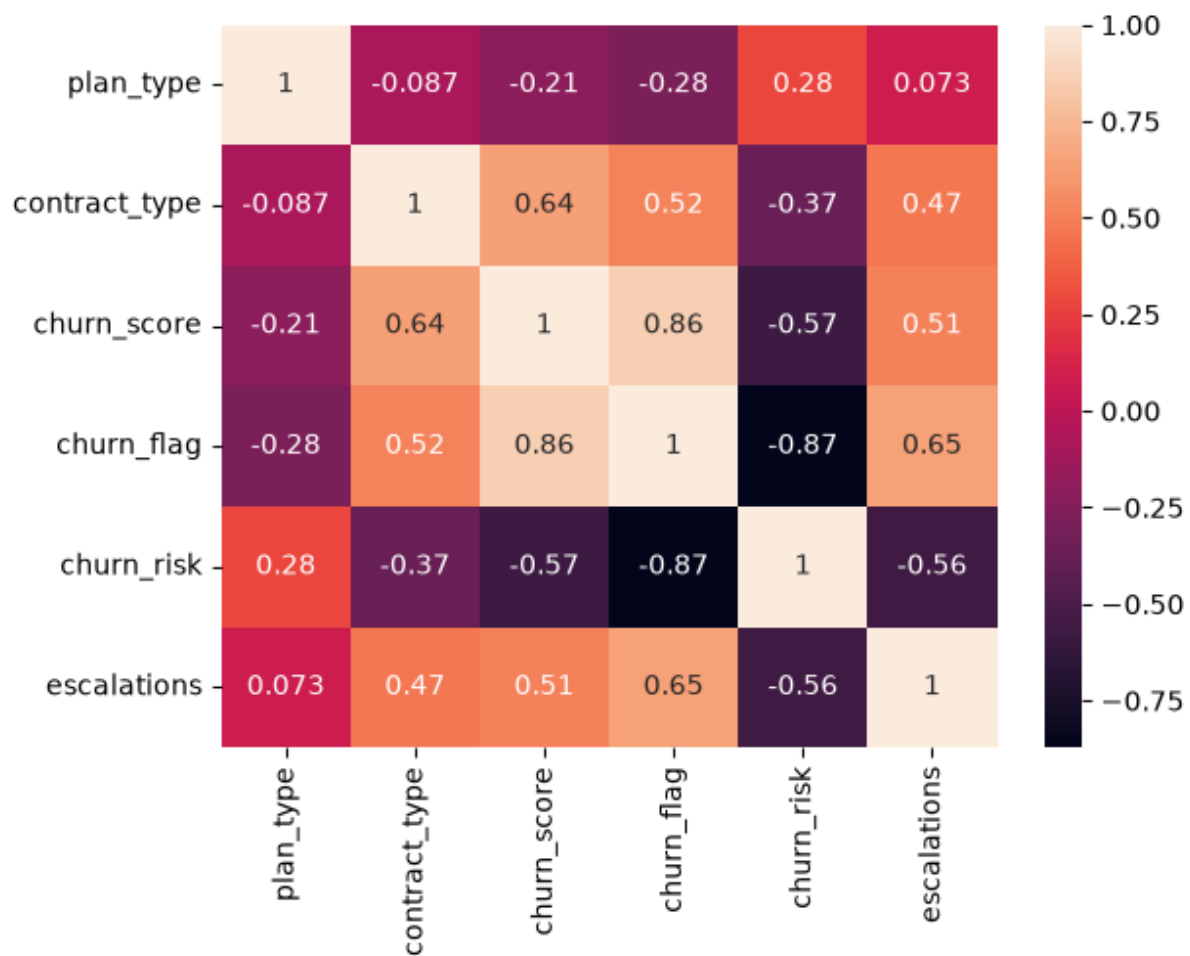
17.2 Churn by Plan Type

This bar chart compares the average churn_flag across Basic, Premium and Standard plans.



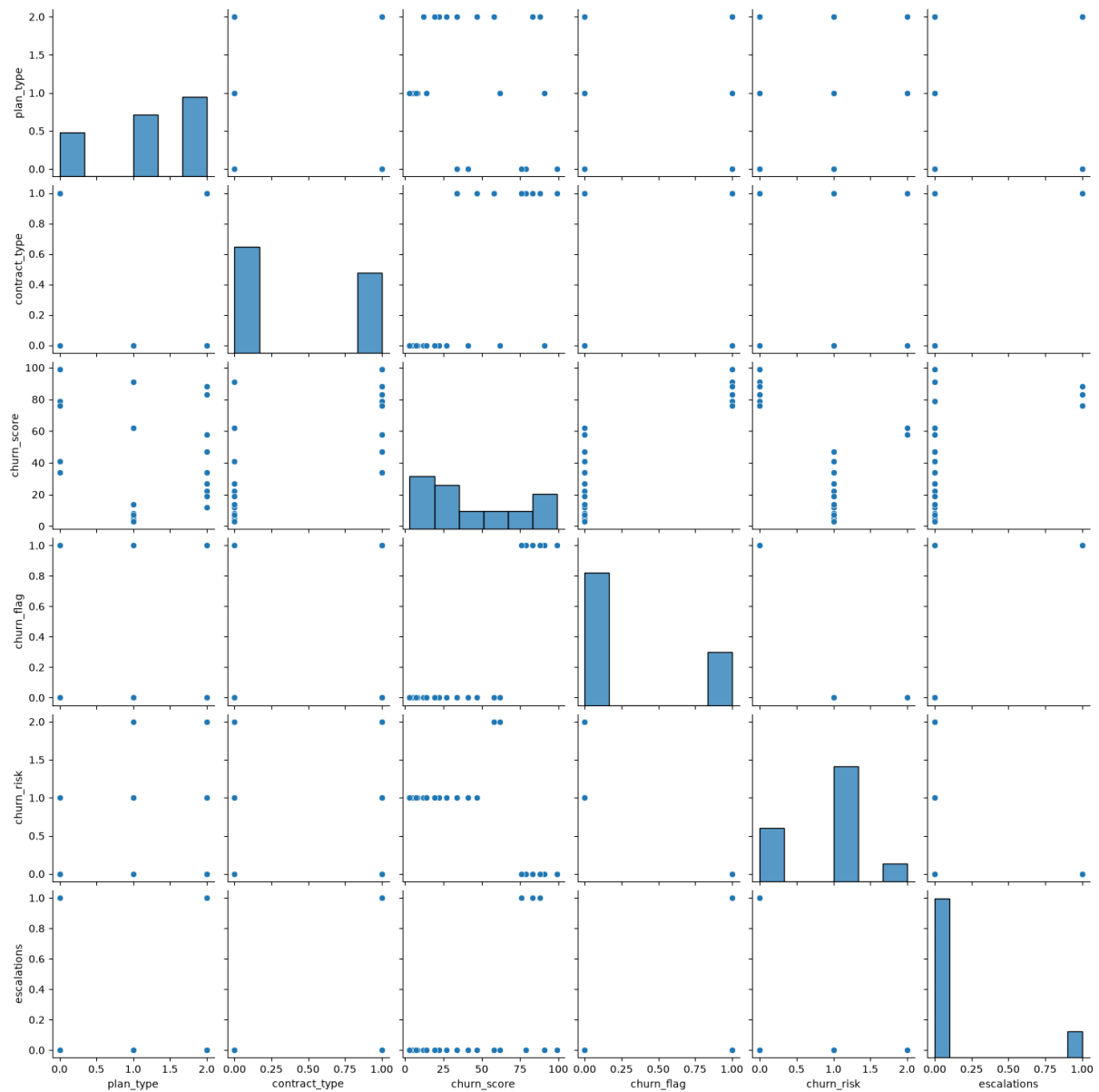
17.3 Correlation Heatmap

The Seaborn heatmap displays correlations among the encoded plan type, contract type, churn score, churn flag, churn risk and escalation features.



17.4 Pairplot

The pairplot provides a multidimensional view of relationships among the encoded numerical features used in the notebook.



17.5 Multidimensional Categorical Analysis

The Seaborn categorical plot compares monthly charges by plan type, gender and churn-risk category.

